KNOT GRAPHS AND GROMOV HYPERBOLICITY

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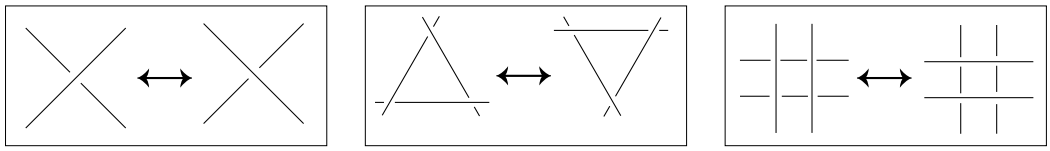
Abstract. We define three types of knot graphs, constructed with the help of unknotting operations, the concordance relation and knot invariants. Some of these graphs have been previously studied in the literature, others are defined here for the first time. The main question we pose and answer in a host of cases is whether the knot graphs are Gromov hyperbolic. With the exception of a particular family of quotient knot graphs, we find that overwhelmingly, they are not.

1. Introduction

1.1. **Background and Motivation.** Recent years have seen a substantial increase of interest in Gromov hyperbolicity on metric spaces. Gromov's definition of hyperbolic groups [Gro87] and subsequent work found a vast array of applications within and beyond mathematics. Hyperbolic groups have been found to be relevant to automatic groups, which play a role in the science of computing, and hyperbolic groups have been applied to secure transmission of information over the internet, to the spread of viruses over networks, and even in the study of DNA molecules. See [BKM01, JL02, JL04, Ohs02] and the references therein.

Mathematicians have found hyperbolic groups important in the study of hyperbolic geometry, low-dimensional topology and combinatorial group theory. These groups satisfy nice algebraic properties such as the Tits alternative, have geometric properties like exponential growth and satisfy linear isoperimetric inequlities, and have homological properties facilitating, for example, a comparison between bounded and ordinary cohomology, see [BFH00, Bow06, Min02, Tit72] for more details.

An interesting result by Bowditch [Bow91] (see also Chapter 6 in [Gro87] as well as [PRT04, RT04, RT07, Tou11]) posits that hyperbolicity of a geodesic metric space is equivalent to the hyperbolicity of a graph associated to it, underscoring the "approximately-tree-like" nature of hyperbolic spaces. This result puts the onus on understanding and exploring hyperbolicity in graphs, which is partially the motivation for this work. Indeed, knot theory provides a bountiful source of graphs, naturally constructed with the aid of knot invariants and/or unknotting operations on knot diagrams. The study of these graphs and their properties have been explored rather poorly thus far (see however [GG16, HU02, IJ11, NO09, Ohy06, ZYL17,



(a) (b) (c)

FIGURE 1. Examples of unknotting operations: (a) Crossing change, (b) Delta-move (or Δ-move), (c) Pass move (or #-move).

ZY18]). It is our wish to shed light on these objects, particularly on their hyperbolicity properties, but also others. To describe these graphs, here are first some preliminaries.

Let (X, d) be a geodesic metric space (see Section 2 for details). For $\delta \geq 0$ we say that a geodesic triangle with edges $\{\alpha, \beta, \gamma\}$ is *δ-thin* if each edge is contained in the closed δ-neighborhood of the union of the remaining two. We say that X is *δ-hyperbolic* if every geodesic triangle in X is δ-thin, and we call X *(Gromov) hyperbolic* if it is δ-hyperbolic for some $\delta \geq 0$.

A rich source of examples of geodesic metric spaces comes from connected graphs. Indeed, a connected graph G can be viewed as a geodesic metric space in which the distance between a pair of vertices is the minimum number of edges in G needed to create a "path" from one vertex to the other (see again Section 2 for more details). Given any connected (infinite) graph G, it is meaningful then to ask whether or not its "edge-metric" is Gromov hyperbolic. We take this question and ask it for the knot graphs alluded to above.

1.2. **Hyperbolicity in Knot Graphs.** Recall that an *unknotting operation $\mathcal{O}$ on knot diagrams* is a local modification/move on a knot diagram, with the property that any knot diagram can be unknotted with a finite number of such $\mathcal{O}$-moves or their inverses. Examples of unknotting operations abound and include the crossing change operation, the Delta-move, the pass move (all three shown in Figure 1), and the infinite family of $H(n)$-moves, $n \geq 2$ from Figure 2. Note that many of these operations may be defined independent of any particular projection of the knot.

Definition 1.1. For $n \in \mathbb{N}$, let $\mathcal{O}_1, \ldots, \mathcal{O}_n$ be n distinct unknotting operations on knot diagrams, and let $\mathbb{O} = \{\mathcal{O}_1, \ldots, \mathcal{O}_n\}$. For $m \in \mathbb{N}$ let $\mathcal{I}_1, \ldots, \mathcal{I}_m$ be integer-valued knot invariants and let $\mathbb{I} = \{\mathcal{I}_1, \ldots, \mathcal{I}_m\}$.

(i) The *$\mathbb{O}$-Gordian Knot Graph* $\mathcal{K}_{\mathbb{O}}$ associated to the collection of unknotting operations $\mathbb{O}$ is the graph whose vertices are unoriented knots, and in which a pair of knots K and K' span an edge if they possess diagrams related by an $\mathcal{O}_i$-move (or its inverse) for some $i \in \{1, \ldots, n\}$.

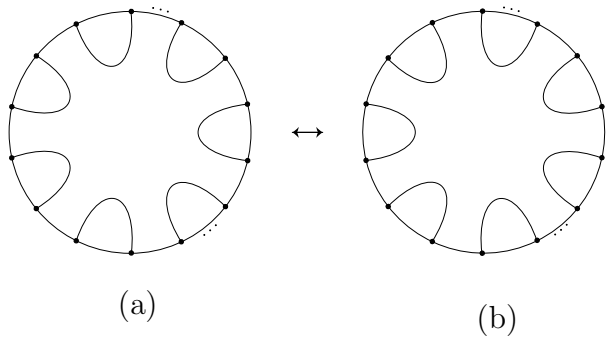


FIGURE 2. The $H(n)$-move is the unknotting operation which replaces the pattern from subfigure (a) in a knot diagram, with the pattern from subfigure (b). The operation is to be performed so as to preserve the number of components. Shown here is the example of $n = 7$. These moves were first introduced and studied by Hoste, Nakanishi and Taniyama [HNT90].

(ii) The *Concordance Knot Graph* $\mathcal{CK}_{\mathbb{O}}$ associated to the collection of unknotting operations $\mathbb{O}$ is the graph whose vertices are concordance classes $[K]$ of oriented knots K, and in which a pair of concordance classes $[K]$ and $[K']$ span an edge if there exist oriented knots L and L' concordant to K and K' respectively, and such that L and L' possess diagrams related by an $\mathcal{O}_i$-move (or its inverse) for some $i \in \{1, \ldots, n\}$.

(iii) The *Quotient Knot Graph* $\mathcal{QK}^{\mathbb{I}}_{\mathbb{O}}$ associated to the collection of unknotting operations $\mathbb{O}$ and the collection of knot invariants $\mathbb{I}$ is the graph whose vertices are equivalence classes $[K]^{\mathbb{I}}_{\mathbb{O}}$ of knots K, by which a pair of knots K and K' are equivalent if $\mathcal{I}_i(K) = \mathcal{I}_i(K')$ for all $i = 1, \ldots, m$. Two equivalence classes $[K]^{\mathbb{I}}_{\mathbb{O}}$ and $[K']^{\mathbb{I}}_{\mathbb{O}}$ span an edge if there exist knots L and L' equivalent to K and K' respectively, and such that L and L' possess diagrams related by an $\mathcal{O}_i$-move (or its inverse) for some $i \in \{1, \ldots, n\}$.

We shall collectively refer to these 3 types of graphs as *Knot Graphs*.

Remark 1.1. In part (iii) of the definition above, we assume that all invariants $\mathcal{I}_i$ in the collection $\mathbb{I}$ are preserved under orientation reversal.

Remark 1.2. Following [HU02] one can define the structure of a simplicial complex on all the knot graphs, by letting a collection of $n + 1$ vertices span an n-simplex if each pair of vertices spans an edge. This leads to very rich simplicial structures on the knot graphs. For example, it has been shown that in the knot graphs $\mathcal{K}_{\times}$, $\mathcal{K}_{H(n)}$,

$\mathcal{K}_{\#}$, $\mathcal{K}_{RCC}$ ("RCC" = Region Crossing Change) each edge of the graph lies in an n-simplex for any $n \in \mathbb{N}$, cf.[HU02, ZYL17, ZY18, GPV19] respectively.

To the best of our knowledge, Definition 1.1 is sufficiently general to include all instances of knot graphs that have appeared in the literature thus far. The knot graph $\mathcal{K}_{\times}$ (where $\times$ indicates the crossing change operation) is referred to as the *Gordian graph*, and has been studied for instance in [BCJ+17, GG16, HU02]. The *band-Gordian graph* $\mathcal{K}_{H(2)}$ and its analogues, the $H(n)$*-Gordian graphs* for $n \geq 3$, have been considered in [ZYL17], and the *pass-move Gordian complex* $\mathcal{K}_{\#}$ appears in [NO09]. We are not aware of any previous results about the concordance knot graphs $\mathcal{CK}_{\mathbb{O}}$, but [IJ11] studies $Q\mathcal{K}^{\mathcal{I}}_{\mathcal{O}}$ with $\mathcal{I}$ the Conway polynomial and $\mathcal{O}$ the pass-move, while [Ohy06] studies $Q\mathcal{K}^{\mathbb{I}}_{\mathbb{O}}$ with $\mathbb{I}$ being the Vassiliev invariants and $\mathbb{O}$ being the collection of C_k-moves.

Different choices for $m, n, \mathcal{O}_i$ and $\mathcal{I}_j$ lead to infinitely many examples of knot graphs. In this work we restrict ourselves to these choices:

$$\text{Unknotting Operations} = \begin{cases} \times & = \text{ Crossing change} \\ H(n) & = \text{ The } H(n)\text{-move from [HNT90]} \end{cases}$$

$$\text{Knot Invariants} = \begin{cases} g_4 & = \text{ Orientable smooth 4-genus} \\ \gamma_4 & = \text{ Nonorientable smooth 4-genus} \\ u & = \text{ Unknotting number} \\ \tau & = \text{ Ozsváth-Szabó's tau invariant} \\ s & = \text{ Rasmussen's } s \text{ invariant} \end{cases}$$

It should be noted that many unknotting operations can be realized as one of the $H(n)$-moves. For example, both the crossing change move and the #-move from Figure 1 can be realized as $H(3)$-moves.

Definition 1.2. An unknotting operation $\mathcal{O}$ and an integer valued knot invariant $\mathcal{I}$ are said to be *compatible* if changing a knot K by a single $\mathcal{O}$-move (or its inverse) changes $\mathcal{I}(K)$ by at most 1. Said differently, if K and K' are knots related by an $\mathcal{O}$-move or its inverse, then $|\mathcal{I}(K) - \mathcal{I}(K')| \leq 1$.

A quotient graph $Q\mathcal{K}^{\mathbb{I}}_{\mathbb{O}}$ (Part (iii) of Definition 1.1) is said to be *compatible* if every $\mathcal{O}_i \in \mathbb{O}$ is compatible with every $\mathcal{I}_j \in \mathbb{I}$.

Note that certain knot invariants differ by a universally bounded amount other than one (for example the signature differs by at most two under a crossing change). For such invariants, one may scale the invariant in order to obtain an appropriately compatible knot invariant.

Our results below about quotient knot graphs $Q\mathcal{K}^{\mathbb{I}}_{\mathbb{O}}$ only apply to the compatible variety. Non-compatible quotient graphs can be rather exotic, see Example 5.6. Our main result shows that hyperbolicity is not naturally inherent in knot graphs, with

the exception of those compatible quotient graphs where $\mathbb{I}$ contains only a single knot invariant $\mathcal{I}$.

Theorem 1.3. *For any $\delta \geq 0$ there exists a geodesic triangle that is not δ-thin in*

(i) *the knot graphs $\mathcal{K}_{\times}$ and $\mathcal{K}_{H(n)}$, for all $n \geq 2$.*
(ii) *the concordance graph $\mathcal{CK}_{\times}$.*
(iii) *the quotient knot graph $Q\mathcal{K}_{\times}^{\{g_4,u\}}$.*

Accordingly, these graphs are not δ-hyperbolic for any $\delta \geq 0$, and therefore not Gromov hyperbolic.

The result about the Gordian graph $\mathcal{K}_{\times}$ from Part (i) of this theorem is alluded to in work of Gambaudo and Ghys [GG16]. In particular, they establish a quasi-isometric embedding of the integer lattice $\mathbb{Z}^d$ into the set of knots with a metric equivalent to the edge-metric on the knot graph $\mathcal{K}_{\times}$. They also note the naturality of this metric in the sense that the Gordian distance between two knots is the minimum number of generic double points over immersed homotopies relating them.

In contrast to the preceding theorem, we have this hyperbolicity result.

Theorem 1.4. *Each of the quotient knot graphs*

$$Q\mathcal{K}_{\times}^{g_4}, \qquad Q\mathcal{K}_{\times}^{u}, \qquad Q\mathcal{K}_{H(2)}^{\gamma_4}, \qquad Q\mathcal{K}_{\times}^{\tau}, \qquad \text{and} \qquad Q\mathcal{K}_{\times}^{s/2},$$

is δ-hyperbolic for any $\delta \geq 0$. Specifically, the first three spaces are isometric to $\mathbb{N} \cup \{0\}$, while the second two are isometric to $\mathbb{Z}$, both equipped with the Euclidean metric.

Remark 1.3. The results presented in Theorems 1.3 and 1.4 stand in stark contrast to one another, representing opposite extremes on the "δ-hyperbolicity scale". It would appear that hyperbolicity in quotient knot graphs emerges only when the set of invariants $\mathbb{I}$ used in its construction consists of a single knot invariant, and when that knot invariant is compatible with all the unknotting operations in $\mathbb{O}$. Indeed, in such a case we find the resulting quotient knot graph to be quasi-isometric to a subset of $\mathbb{R}$ (cf. Theorem 5.1). In all other cases we find that hyperbolicity is absent from knot graphs.

Question 1.5. *Does there exist a knot graph that is δ-hyperbolic for some, but not all $\delta > 0$?*

1.3. **Homogeneity and Links in Knot Graphs.** A metric space (X, d) is *homogeneous* if for every $x, y \in X$ there exists an isometry $\psi : X \to X$ with $\psi(x) = y$, i.e. if the isometry group of X acts transitively on X. If a metric space (G, d) arises from a graph G all of whose edges have length 1, let us define the *link of a vertex* $v \in Vert(G)$, denoted $\ell k(v)$, as the induced subgraph of G generated by the set

$$\{w \in Vert(G) \,|\, d(v, w) = 1\}.$$

Note that $v \notin \ell k(v)$ and that for $w, u \in \ell k(v)$, an edge $e = \{w, u\}$ belongs to $\ell k(v)$ if and only if $d(w, u) = 1$. The *diameter of* $\ell k(v)$ is the supremum of $\{d(w, u) \mid w, u \in Vert(\ell k(v))\}$. If (G, d) is a homogeneous metric space, clearly the links of any pair of vertices are isometric.

Question 1.6. *With regards to the above definition, we ask:*

(i) *In which, if any, knot graphs is the link of the (class of the) unknot connected?*
(ii) *If the link of the (class of the) unknot is connected, determine if its diameter is finite. If the diameter is finite, calculate or estimate its value.*
(iii) *Which, if any, knot graphs from Definition 1.1 are homogeneous?*

Some of these questions are inspired by the work [HW15] of Hoffman-Walsh which studies the *Big Dehn Surgery Graph*. The vertices of this graph are closed orientable 3-manifolds, and edges are formed by 3-manifolds related by a Dehn surgery. Hoffman-Walsh prove that the link of S^3 is connected and of finite diameter.

In another direction, Nakanishi and Ohyama [NO09] show that the #-Gordian graph $\mathcal{K}_{\#}$ is not homogeneous, by utilizing the well understood relation between the Conway polynomial and pass-moves.

Theorem 1.7. *Let $\mathbb{O} = \{\mathcal{O}_1, \ldots, \mathcal{O}_n\}$ be a collection of unknotting operations and let $\mathcal{CK}_{\mathbb{O}}$ be the associated concordance graph.*

(i) *The concordance graph $\mathcal{CK}_{\mathbb{O}}$ is always homogeneous. Specifically, an isometry of $\mathcal{CK}_{\mathbb{O}}$ sending a concordance class $[K]$ to a concordance class $[K']$ is given by*
$$\psi([L]) = [L\#(-\overline{K})\#K'],$$
where $-\overline{K}$ is the reverse mirror of K.
(ii) *The quotient knot graphs $\mathcal{QK}^{\tau}_{\times}$ and $\mathcal{QK}^{s'}_{\times}$ are homogeneous.*

Part (ii) of the preceding theorem is a direct consequence of Theorem 1.4.

Proposition 1.8. *The link of the class of the unknot in the quotient knot graphs $\mathcal{QK}^{g_4}_{\times}$, $\mathcal{QK}^{u}_{\times}$ and $\mathcal{QK}^{\gamma_4}_{H(2)}$ is a singleton set. In the quotient knot graphs $\mathcal{QK}^{\tau}_{\times}$ $\mathcal{QK}^{s/2}_{\times}$, the link of the class of the unknot consists of exactly two points and is disconnected.*

1.4. **Organization.** In Section 2, we discuss the metrics on graphs and geodesics in the resulting metric spaces. We also review the Gromov hyperbolicity for geodesic metric spaces, and definitions and properties of quasi-isometries. In Subsection 2.4, we review the properties of the induced metric on the $H(n)$-Gordian graph, proving that the $H(n)$-Gordian graph and the $H(m)$-Gordian graph are quasi-isometric for any $m, n \geq 2$. In Subsection 2.5, we discuss the bounds on the induced metric on the $H(n)$-Gordian graph coming from cyclic branched covers. At the end of Section 2, we review the algorithm for computing the homology groups of Brieskorn manifolds,

following Orlik [Orl72]. In Section 3, we construct a geodesic triangle which is not δ-thin for any $\delta \geq 0$ in the $H(n)$-Gordian graph, proving part (i) of Theorem 1.3. In Section 4, we prove part (ii) of Theorem 4 and deduce that the classical Gordian graph is also not Gromov hyperbolic. We also give the proof of part (i) of Theorem 1.7 there. In Section 5, we prove part (iii) of Theorem 1.3 and Theorem 1.4.

2. Background Material

2.1. **Geodesic Metric Spaces.** Let (X, d) be a metric space, and let $\alpha : [a, b] \to X$ be a path. Given a partition $\mathcal{P} = \{t_0, \ldots, t_n\}$ of $[a, b]$, let

$$L(\alpha, \mathcal{P}) = \sum_{i=1}^{n} d(\alpha(t_{i-1}), \alpha(t_i))$$

denote the *polygonal length of* α *asssociated to the partition* $\mathcal{P}$. We say that α is a *rectifiable path* if the supremum of its polygonal lengths, taken over all partitions of $[a, b]$, is finite. In that case we define the *length* $L(\alpha)$ *of* α as said supremum:

$$L(\alpha) = \sup_{\mathcal{P}} L(\alpha, \mathcal{P}).$$

It is easy to check that if $\alpha : [a, b] \to X$ is a rectifiable path, then so is its restriction to any segment $[c, d] \subset [a, b]$.

A metric space (X, d) is called *geodesic metric space* if for every pair of points $x, y \in X$ there exists a rectifiable path $\alpha : [0, 1] \to X$ with $\alpha(0) = x$, $\alpha(1) = y$ and with

$$L(\alpha|_{[s,t]}) = d(\alpha(s), \alpha(t)), \qquad \forall s, t \in [0, 1].$$

Any such path α is called a *geodesic path.* A *geodesic triangle* $\{\alpha, \beta, \gamma\}$ in a geodesic metric space (X, d) is a triple of geodesics $\alpha, \beta, \gamma : [0, 1] \to X$ with $\alpha(1) = \beta(0)$, $\beta(1) = \gamma(0)$ and $\gamma(1) = \alpha(0)$. We refer to α, β and γ (or sometimes their images in X) as the *edges* of the geodesic triangle, and the points $\{\alpha(1), \beta(1), \gamma(1)\}$ as the *vertices* of the geodesic triangle.

Recall from the introduction that for $\delta \geq 0$, a geodesic triangle $\{\alpha_1, \alpha_2, \alpha_3\}$ is called δ-thin if for every $i \in \{1, 2, 3\}$ and every $x \in \mathrm{Im}(\alpha_i)$, the inequality

$$d(x, \cup_{j \neq i} \mathrm{Im}(\alpha_j)) \leq \delta$$

holds. The geodesic metric space (X, d) is called δ-hyperbolic if every geodesic triangle in X is δ-thin, and we say that (X, d) is Gromov hyperbolic if it is δ-hyperbolic for at least one $\delta \geq 0$. Observe that if X is δ-hyperbolic then it is also δ'-hyperbolic for every $\delta' \geq \delta$. If X is Gromov hyperbolic, we let

$$\delta(X) = \inf\{\delta \geq 0 \,|\, X \text{ is } \delta\text{-hyperbolic}\}.$$

2.2. **Graphs as Geodesic Metric Spaces.** Let G be a graph and let $Vert(G)$ and $Edge(G)$ denote its sets of vertices and edges respectively. A graph G can be viewed as a 1-dimensional CW complex whose 0-cells are the vertices of G, and whose 1-cells are in one-to-one correspondence with the edges of G. Specifically, for each edge $e \in Edge(G)$ with endpoints $v, w \in Vert(G)$ we attach a 1-cell $\alpha_e \cong [0,1]$ to $Vert(G)$ whose attaching map identifies the two endpoints $\{0,1\}$ of the 1-cell α_e with v and w. This endows the graph G with the structure of a topological space, in such a way that G is connected as a graph if and only if it is path-connected as a topological space.

We next define a metric d on a connected graph G, by first defining it for vertices $v, w \in Vert(G)$ as:

$$d(v,w) = \text{ Minimum number of edges needed to connect } v \text{ to } w.$$

Note that this definition tacitly gives each edge in the graph length 1. The distance between a pair of points x, y lying on the same 1-cell $\alpha_e \cong [0,1]$ is

$$d(x,y) = \begin{cases} |x-y| & ; \quad \text{if } \alpha_e \text{ has two distinct endpoints,} \\ \min\{|x-y|, |x|+|1-y|\} & ; \quad \text{if } \alpha_e \text{ has only one endpoint.} \end{cases}$$

In the above definition, we assume in the second case that the attaching map takes the unit interval to a circle of radius $1/2\pi$ so that $d(x,y)$ just corresponds with the distance along a circle of circumference one. Lastly, given points x, y lying on distinct edges α_e and $\alpha_{e'}$ with boundary vertices $\{v_0, v_1\}$ and $\{w_0, w_1\}$ respectively, we define their distance $d(x,y)$ as

$$d(x,y) = \min_{i,j \in \{0,1\}} d(x, v_i) + d(v_i, w_j) + d(w_j, y).$$

With these definitions in place, it is now easy to verify that for a connected graph G, the pair (G,d) becomes a geodesic metric space. We shall use this structure on graphs implicitly on all knot graphs in subsequent sections.

2.3. **Quasi-isometries and hyperbolicity.** A map $f : X_1 \to X_2$ between metric spaces $(X_1, d_1), (X_2, d_2)$ is called a *quasi-isometric embedding* if there are constants $a \geq 1, b \geq 0$ such that the double inequality

$$\frac{1}{a} d_1(x,x') - b \leq d_2(f(x), f(x')) \leq a d_1(x,x') + b,$$

holds for all $x, x' \in X_1$. In addition, if there is a constant $C \geq 0$ such that for every $y \in Y$ there exists an $x \in X$ with

$$d_2(y, f(x)) \leq C,$$

then X_1 and X_2 are called *quasi-isometric*. If $C = 0$, the map f is called *bi-Lipschitz*.

Gromov hyperbolicity is invariant under quasi-isometries between geodesic spaces.

Proposition 2.1. [Ghy90, Theorem 12] *Assume that (X_1, d_1) is quasi-isometric to (X_2, d_2) with parameters a, b and C. If X_1 is δ-hyperbolic, then X_2 is δ'-hyperbolic with δ' depending on δ, a, b, C.*

Corollary 2.2. *If (X_1, d_1) and (X_2, d_2) are quasi-isometric geodesic metric spaces and X_1 is not δ-hyperbolic for any $\delta \geq 0$, then neither is X_2.*

2.4. **$H(n)$-moves.** The $H(n)$-move, $n \geq 2$, is defined in Figure 2. We adopt the convention from [HNT90] that only those $H(n)$-moves are allowed which preserve the number of components. The $H(2)$-move is called a *noncoherent* or *nonorientable band move*, as it is realized by attaching a band to the knot, in such a way that the orientation of the band agrees with that of the knot at one of its ends, and disagrees at the other.

The $H(n)$-moves were introduced and studied by Hoste, Nakanishi and Taniyama in [HNT90], where they proved that each $H(n)$-move is an unknotting operation. We are thus justified in letting $\mathcal{K}_{H(n)}$ denote the resulting $H(n)$-Gordian knot graphs, and we denote the induced metric on $\mathcal{K}_{H(n)}$ by d_n. Hoste, Nakanishi and Taniyama established several estimates for the $H(n)$-unknotting number $u_n(K)$, defined as $d_n(K, U)$ (with U the unknot). The following theorem is proved in [HNT90] for the case of $K' = U$, we adapt their proofs for our somewhat more general formulas.

Theorem 2.3 (Hoste, Nakanishi, Taniyama [HNT90])**.** *Let K, K' be a pair of knots and $n \geq 2$ an integer.*

(i) *An $H(n)$-move can be realized by an $H(n+1)$-move. In particular*

$$d_n(K, K') \geq d_{n+1}(K, K').$$

(ii) $\lim_{n\to\infty} d_n(K, K') = 1$.

(iii) *If $n \geq 3$ then $(n-1)d_n(K, K') \geq \frac{2}{3}d_2(K, K') + 1$.*

Proof. (i) The fact that an $H(n)$-move can be realized as an $H(n+1)$-move is shown in Lemma 2 and Figure 10 in [HNT90]. From this the inequality $d_n(K, K') \geq d_{n+1}(K, K')$ is obvious.

(ii) For $K' = U$ this formula is the content of Theorem 6 in [HNT90]. We modify the proof of the said theorem to obtain the claimed result. Each $H(n)$-move can be obtained by a sequence of $(n-1)$ $H(2)$-moves, each of which is realized by attaching a noncoherent band as in Figure 3. Let $n \geq 2$ be arbitrary. Since one can pass from a diagram for K to a diagram for K' by applying $d_n(K, K')$ $H(n)$-moves, it follows that the diagrams of K and K' are related by $(n-1) \cdot d_n(K, K')$ noncoherent band attachments. By sliding bands if necessary, we may assume that all the bands are disjoint. Furthermore, we may gather the root of each band near one point of the knot K' as in Figure 4. It is now an easy observation that all $(n-1)d_n(K, K')$

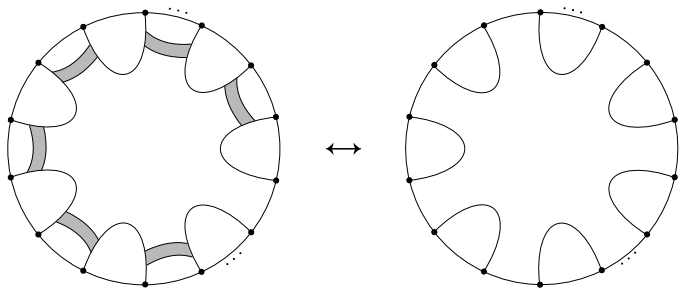


FIGURE 3. An $H(n)$-move can be realized by $n-1$ $H(2)$-moves, i.e. by attaching $n-1$ noncoherent bands, as indicated. Pictured here is the case of $n = 7$.

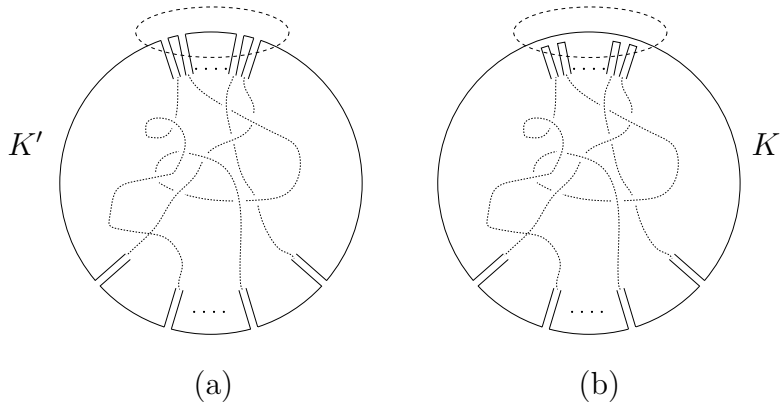


FIGURE 4. If a knot K' can be obtained from K by m $H(n)$-moves, then K' can also be obtained from K by $m(n-1)$ $H(2)$-moves, each of which is realized by the attaching of a noncoherent band. If the roots of the bands are gathered as shown, the totality of all $m(n-1)$ band moves is accomplished by a single $H(m(n-1))$-move, the one inside the dashed oval.

noncoherent band moves are realized by a single $H((n-1)d_n(K,K'))$-move (see again Figure 4), showing that

$$d_{(n-1)d_n(K,K')}(K,K') = 1.$$

The proof of Part (ii) of the present theorem follows from this formula and Part (i).

(iii) This formula for $K' = U$ is Part (6) of Theorem 7 in [HNT90], and the proof is readily adapted for our purposes. Let $m \geq 2$ be such that $d_m(K,K') = 1$ (such an

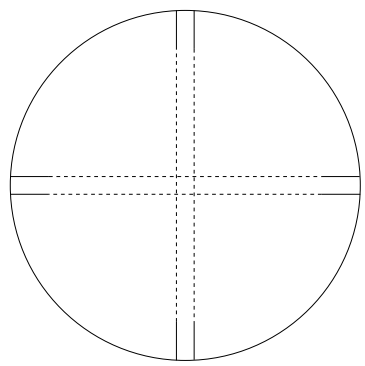


FIGURE 5. A special pair of bands used to related the diagrams of K and K'.

m exists by Part (ii), e.g. $m = (n-1)d_n(K, K')$ for any $n \geq 2$). Then the diagrams of K and K' can be related by a single $H(m)$-move, and hence also by $m-1$ band attachments. Individually, each of these $m-1$ band attachments may be component preserving or may change the number of components by one. In particular, cutting one band either yields another knot or a two-component link. Each band of the first kind can be removed by a single $H(2)$-move. Consider then a band of the second variety, and specifically consider an "inner-most"one, i.e. a band whose roots divide the knot into two arcs, each of which contains at most one root of any other band. It must be then that there exists a pair of bands as in Figure 5. It is shown in [HNT90], Figure 15, that this pair of bands can be removed by 3 $H(2)$-moves. Therefore, we can remove noncoherent bands with either a single $H(2)$-move, or we can remove pairs of noncoherent bands with 3 $H(2)$-moves. Since removing all $m-1$ noncoherent bands changes a diagram for K to one for K', we obtain the inequality

$$\frac{3}{2}(m-1) \geq d_2(K, K').$$

The claim now follows since we may pick $m = (n-1)d_n(K, K')$ for any $n \geq 3$. □

A direct and important consequence of the preceding theorem is this observation.

Corollary 2.4. *The $H(n)$-Gordian graph $\mathcal{K}_{H(n)}$ for $n \geq 3$ is quasi-isometric to the $H(2)$-Gordian graph $\mathcal{K}_{H(2)}$.*

2.5. **Bounds on d_n coming from cyclic branched covers.** For a knot L, let $\Sigma_m(L)$ denote the closed 3-manifold obtained as the m-fold cyclic cover of S^3 with branching set the knot L. If $m = 2$ we simply write $\Sigma(L)$ instead of $\Sigma_2(L)$. Let $e_m(L)$ denote the minimum number of generators of $H_1(\Sigma_m(L); \mathbb{Z})$ and let $e_m^p(L)$

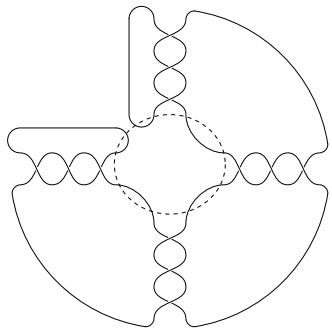


FIGURE 6. A 4-fold connected sum of $T(3,2)$ with itself, can be unknotted by the single $H(5)$-move indicated in the dashed oval. A generalization of this picutre shows that the connected sum $\#^{n-1}T(2,k)$ for any odd k and $n \geq 2$, can be unknotted by a single $H(n)$-move.

denote the minimum number of generators of $H_1(\Sigma_m(L);\mathbb{Z}_p)$ (here and elsewhere $\mathbb{Z}_p$ denotes the cyclic group on p elements).

Proposition 2.5. [HNT90, Theorem 4] *Let K and K' be a pair of knots and let $m, n \geq 2$, then*

$$\frac{|e_m(K) - e_m(K')|}{(n-1)(m-1)} \leq d_n(K, K').$$

Proof. The case of $K' = U$ is Theorem 4 in [HNT90], and we use its proof as a starting point for the proof presented here. Suppose a single $H(n)$-move changes a knot L to a knot L'. Then $\Sigma_m(L)$ and $\Sigma_m(L')$ are related by a surgery on a handlebody of genus $(m-1)(n-1)$ (as the m-fold branched cover of a "thickened disk"containing the loops as in Figure 2 is a handlebody of said genus). Lemma 3 from [HNT90] shows that in this case the inequality $|e_m(L) - e_m(L')| \leq (m-1)(n-1)$ holds.

If K' is obtained from K by $d_n(K, K')$ $H(n)$-moves, with intermediate knots

$$K = K_0 \to K_1 \to \cdots \to K_{d_n(K,K')} = K',$$

then

$$|e_m(K) - e_m(K')| \leq \sum_{i=1}^{d_n(K,K')} |e_m(K_{i-1}) - e_m(K_i)| \leq d_n(K, K')(m-1)(n-1),$$

proving the claim. □

Recall that the invariants $e_2^3(K)$ and $e_2^5(K)$ are the minimum number of generators of the first homology of the branched double cover with coefficients in $\mathbb{Z}/3$ and $\mathbb{Z}/5$, respectively. As it turns out, these invariants are related to certain evaluations of the Jones and Q-polynomials (see [LM86, Jon89]). In [AK14], Abe-Kanenobu give criteria for knots to be related by an $H(2)$-move in terms of evaluations of these polynomials. This can in turn be rephrased as a lower bound in terms of the invariants $e_2^3(K)$ and $e_2^5(K)$ as follows:

Proposition 2.6. [AK14, Corollary 5.6, Corollary 5.10] *Let K and K' be a pair knots. Then*

1. $|e_2^3(K) - e_2^3(K')| \leq d_2(K, K')$, *and*
2. $|e_2^5(K) - e_2^5(K')| \leq d_2(K, K')$.

2.6. **Homology of Brieskorn manifolds.** In this section, we review the algorithm for computing the homology groups of Brieskorn manifolds, following Orlik [Orl72].

For integers $w_1, w_2, w_3 > 1$,the Brieskorn manifold $\Sigma(w_1, w_2, w_3)$ is defined as

$$\{(z_1, z_2, z_3) \in \mathbb{C}^3 \mid z_1^{w_1} + z_2^{w_1} + z_3^{w_1} = 0\} \cap S_\varepsilon$$

where S_ϵ is a sphere in $\mathbb{C}^3$ centered at 0 and of radius $\epsilon > 0$ chosen sufficiently small so that the only singularity contained inside of S_ϵ is at $z = 0$. Brieskorn manifolds are closed, oriented 3-manifolds, and by Milnor [Mil75], they can also be obtained as an r-fold cyclic cover of S^3 with the branching set the torus knot or link $T(p, q)$.

To determine both the rank and the torsion subgroup of $H_1(\Sigma(w_1, w_2, w_3); \mathbb{Z})$ we first proceed to define several sets of numbers following [Orl72] (see also the reference [Ran75]), specialized here to varieties of only three variables $\{z_1, z_2, z_3\}$. For any ordered subset I of $\{1, 2, 3\}$ let $\kappa(I)$ be defined as

$$\kappa(I) = \sum_{J \subseteq I} (-1)^{|J|-|I|} \frac{\prod w_J}{\mathrm{lcm}(w_J)},$$

where $w_J \subseteq \{w_1, w_2, w_3\}$ is the subset corresponding with indexing subset $J \subseteq \{1, 2, 3\}$. In the definition of $\kappa(I)$ we adopt the convention that $\Pi w_\emptyset / \mathrm{lcm}(w_\emptyset)$ equals 1. Furthermore, let

$$\kappa'(I) = \begin{cases} \kappa(I) & \text{when } |I| \text{ is even, and} \\ 0 & \text{when } |I| \text{ is odd.} \end{cases}$$

Next, define inductively on $|I|$ the numbers $c(I)$ as $c(\emptyset) = 1$ and

$$c(I) = \frac{\gcd(\{w_1, w_2, w_3\} - w_I)}{\prod_{J \subset I} c(J)}.$$

Lastly, let $r = \max_{I \subseteq \{1,2,3\}} \kappa(I)$, and for $j = 1, \ldots, r$ define d_j as

$$d_j = \prod_{I | \kappa'(I) \geq j > 0} c(I).$$

Proposition 2.7. [Orl72, Propositions 2.6 and 3.4] *The rank of $H_1(\Sigma(w_1, w_2, w_3); \mathbb{Z})$ equals $\kappa(\{1, 2, 3\})$ and the torsion subgroup of $H_1(\Sigma(w_1, w_2, w_3); \mathbb{Z})$ is isomorphic to*

$$\mathbb{Z}_{d_1} \oplus \mathbb{Z}_{d_2} \oplus \cdots \oplus \mathbb{Z}_{d_r}.$$

Lemma 2.8. *Let k be an odd natural number, then*

$$H_1(\Sigma(2, k, k); \mathbb{Z}) \cong (\mathbb{Z}_2)^{k-1}.$$

Proof. By Proposition 2.7 the rank of $H_1(\Sigma(2, k, k); \mathbb{Z})$ is zero. A direct computation of $\kappa'(I)$ shows that

$$\kappa'(I) = \begin{cases} 1 & I = \emptyset \\ k-1 & I = \{2, 3\} \\ 0 & \text{otherwise.} \end{cases}$$

The only values of $c(I)$ appearing in the calculation of the d_j are therefore $c(\emptyset) = 1$ and $c(\{2, 3\}) = 2$. Thus $r = k - 1$ and for all $j = 1, \ldots, k - 1$ we obtain

$$d_j = \prod_{\kappa(I) \geq j} c(I) = c(\emptyset) \cdot c(\{2, 3\}) = 2. \qquad \square$$

Example 2.9. Consider the Brieskorn manifold $\Sigma(2, 15, 9)$. Its rank is easily seen to equal zero. Moreover, $\kappa'(\emptyset) = 1$, $\kappa'(\{2, 3\}) = 2$ and $\kappa'(I) = 0$ otherwise. It follows that $r = 2$ and $d_j = c(\emptyset) \cdot c(\{2, 3\}) = 2$ for all $j = 1, 2$. We conclude that

$$H_1(\Sigma(2, 15, 9); \mathbb{Z}) \cong (\mathbb{Z}_2)^2.$$

Example 2.10. The rank of $\Sigma(2, 15, 5)$ is similarly equal zero. Here, $\kappa'(\emptyset) = 1$, $\kappa'(\{2, 3\}) = 4$ and $\kappa'(I) = 0$ otherwise. It follows that $r = 4$ and $d_j = c(\emptyset) \cdot c(\{2, 3\}) = 2$ for all $j = 1, 2, 3, 4$. We conclude that

$$H_1(\Sigma(2, 15, 5); \mathbb{Z}) \cong (\mathbb{Z}_2)^4.$$

3. The $H(n)$-Gordian Knot Graphs

In this section we examine the Gordian knot graphs $\mathcal{K}_{H(n)}$ and prove that for each $\delta \geq 0$ they contain geodesic triangles that are not δ-thin.

Consider any three mutually distinct knots K_0, K_1, K_2. Theorem 2.3 implies that there exists an $n_0 \in \mathbb{N}$ such that for all $n \geq n_0$ the equality $d_n(K_i, K_j) = 1$ holds for any pair of distinct indicies $i, j \in \{0, 1, 2\}$. Accordingly, the knots K_0, K_1, K_2 are the vertices of a geodesic triangle in $\mathcal{K}_{H(n)}$ and this triangle is plainly δ-hyperbolic for all $\delta \geq 1/2$. In contrast to this conclusion, we will show that with $n \geq 2$ fixed,

and for any $\delta \geq 0$, there exist geodesic triangles in $\mathcal{K}_{H(n)}$ which are not δ-hyperbolic. We begin with the case of $n = 2$.

For $m \in \mathbb{N}$, whose value is to be determined later, consider the vertices given by the unknot U, the knot $K_1 = \#^m T(2,9)$ and the knot $K_2 = \#^m T(2,9)\#(\#^m T(2,15))$ in $\mathcal{K}_{H(2)}$. Observe that each of $T(2,9)$ and $T(2,15)$ can be unknotted by a single $H(2)$-move, and hence $d_2(U,K_1) \leq m$ and $d_2(U,K_2) \leq 2m$. Lower bounds on $d_2(U,K_1)$ and $d_2(U,K_2)$ come courtesy of Proposition 2.5. Indeed, since $\Sigma(T(2,9)) \cong L(2,9)$ and $\Sigma(T(2,15)) \cong L(2,15)$, then

$$H_1(\Sigma(K_1);\mathbb{Z}) \cong (\mathbb{Z}_9)^m \quad \text{and} \quad H_1(\Sigma(K_2);\mathbb{Z}) \cong (\mathbb{Z}_9)^m \oplus (\mathbb{Z}_{15})^m .$$

It follows that $e_2(K_1) = m$ and $e_2(K_2) = 2m$ and therefore that $d_2(U,K_1) \geq m$ and $d_2(U,K_2) \geq 2m$, implying that $d_2(U,K_1) = m$ and $d_2(U,K_2) = 2m$. This shows that the path ℓ_1 connecting U to K_1 through the edges with vertices $\#^i T(2,9)$, $i = 1,\ldots,m-1$ is a geodesic path. The same is true of the path ℓ_3 connecting U to K_2 via the edges in $\mathcal{K}_{H(2)}$ with the vertices

$$T(2,15) \leftrightarrow T(2,9)\#T(2,15) \leftrightarrow T(2,9)\#(\#^2 T(2,15)) \leftrightarrow (\#^2 T(2,9))\#(\#^2 T(2,15)) \leftrightarrow \cdots \leftrightarrow (\#^{m-1} T(2,9))\#(\#^{m-1} T(2,15)) \leftrightarrow (\#^{m-1} T(2,9))\#(\#^m T(2,15)).$$

By a similar argument, it follows that $d_2(K_1,K_2) = m$, and the path ℓ_2 connecting K_1 to K_2 passing through the knots $(\#^m T(2,9))\#(\#^i T(2,15))$, $i = 1,\ldots,m-1$, is a geodesic.

Next we shall estimate from below the distance of a particular vertex K_3 from the edge ℓ_3, to the union $\ell_1 \cup \ell_2$ in the geodesic triangle constructed above.

Claim. *Assume that $m = 2k$ is even. Set $K_3 = (\#^k T(2,9))\#(\#^k T(2,15))$, and observe that K_3 is a vertex on the path ℓ_3. Then*

$$d_2(K_3, \ell_1 \cup \ell_2) \geq 3k/4.$$

Proof. We first prove that $d_2(K_3,\ell_1) \geq k$. Let $K = \#^i T(2,9)$ be a vertex on the path ℓ_1, then by Corollary 2.6 we obtain

$$d_2(K_3,K) \geq |e_2^5(K_3) - e_2^5(K)| = k.$$

This concludes that $d_2(K_3,\ell_1) \geq k$.

Now we let K denote a knot on the path ℓ_2 of the form $(\#^{2k} T(2,9))\#(\#^i T(2,15))$ where $i \in \{0,\cdots,2k\}$. Consider the 9-fold cyclic covers of S^3 with branching sets $T(2,9)$ and $T(2,15)$, respectively. These are the Brieskorn manfolds $\Sigma(2,9,9)$ and $\Sigma(2,15,9)$ respectively, and Lemma 2.8 and Example 2.9 imply that

$$H_1(\Sigma(2,9,9);\mathbb{Z}) \cong (\mathbb{Z}_2)^8 \quad \text{and} \quad H_1(\Sigma(2,15,9);\mathbb{Z}) \cong (\mathbb{Z}_2)^2 .$$

Applying Corollary 2.6 we obtain

$$d_2(K_3, K) \geq \frac{|e_9(K) - e_9(K_3)|}{8} = \frac{6k + 2i}{8} \geq \frac{3k}{4}.$$

This implies that $d_2(K_3, \ell_2) \geq 3k/4$ and thus that $d_2(K_3, \ell_1 \cup \ell_2) \geq 3k/4$. Since K_3 lies on ℓ_3, we conclude that ℓ_3 is not contained in the closed δ-neighborhood of $\ell_1 \cup \ell_2$ whenever $3k/4 > \delta$. □

Given any $\delta \geq 0$, choose k in the above construction so that $3k/4 > \delta$, then the geodesic triangle $\{\ell_1, \ell_2, \ell_3\}$ is not δ-thin. It follows that $\mathcal{K}_{H(2)}$ is not δ-hyperbolic for any $\delta \geq 0$.

By Corollary 2.4, $\mathcal{K}_{H(n)}$ is quasi-isometric to $\mathcal{K}_{H(2)}$ for any $n > 2$. Since we already showed that $\mathcal{K}_{H(2)}$ is not δ-hyperbolic for any $\delta \geq 0$, Corollary 2.1 shows that $\mathcal{K}_{H(n)}$ is also not δ-hyperbolic for any $\delta \geq 0$. This proves Part (i) of Theorem 1.3.

We conclude this section with an explicit construction of a geodesic triangle in $\mathcal{K}_{H(n)}$ that can be used to disprove its δ-hyperbolicity directly. Indeed, the construction of said triangle proceeds in analogy with the case $n = 2$ given above. The needed modifications are replacing $T(2, 9)$ by $\#^{n-1}T(2, 9)$ and replacing $T(2, 15)$ by $\#^{n-1}T(2, 15)$. We consider then in $\mathcal{K}_{H(n)}$ the vertices

$$U, \quad K_1 = \#^{m(n-1)}T(2, 9) \quad \text{and} \quad K_2 = (\#^{m(n-1)}T(2, 9))\#(\#^{m(n-1)}T(2, 15)).$$

As before, $m \geq 1$ is an integer whose value is to be determined later. Since each of $\#^{n-1}T(2, 9)$ and $\#^{n-1}T(2, 15)$ can be unknotted by a single $H(n)$-move (as in Figure 6), it follows that $d_n(U, K_1) \leq m$ and $d_n(U, K_2) \leq 2m$. An application of Proposition 2.5 shows that $d_n(U, K_1) \geq m$ and $d_n(U, K_2) \geq 2m$, leading to $d_n(U, K_1) = m$ and $d_n(U, K_2) = 2m$. It follows that the path ℓ_1 connecting U to K_1 by passing through the vertices $\#^{i(n-1)}T(2, 9)$, $i = 1, \ldots, m-1$ is a geodesic path in $\mathcal{K}_{H(n)}$. The same is true of the path ℓ_3 connecting U to K_2 by passing through the vertices

$$\begin{aligned} \#^{(n-1)}T(2, 15) &\leftrightarrow (\#^{(n-1)}T(2, 9))\#(\#^{(n-1)}T(2, 15)) \\ \leftrightarrow (\#^{(n-1)}T(2, 9))\#(\#^{2(n-1)}T(2, 15)) &\leftrightarrow (\#^{2(n-1)}T(2, 9))\#(\#^{2(n-1)}T(2, 15)) \\ \leftrightarrow \cdots \leftrightarrow (\#^{(m-1)(n-1)}T(2, 9))&\#(\#^{(m-1)(n-1)}T(2, 15)) \\ \leftrightarrow (\#^{(m-1)(n-1)}T(2, 9))&\#(\#^{m(n-1)}T(2, 15)). \end{aligned}$$

Lastly let ℓ_2 be the path connecting K_1 to K_2 via edges in $\mathcal{K}_{H(n)}$ with intermediate vertices $(\#^{m(n-1)}T(2, 9))\#(\#^{i(n-1)}T(2, 15))$, $i = 1, \ldots, m-1$. Clearly $d_n(K_1, K_2) \leq m$ (to see this use m $H(n)$-moves to unknot the m summands of $\#^{n-1}T(2, 15)$ in K_2) while an application of Proposition 2.5 gives $d_n(K_1, K_2) \geq m$. This shows that $d_n(K_1, K_2) = m$ and that ℓ_2 is a geodesic path.

As before, let $m = 2k$ with k to be chosen later, and let

$$K_3 = (\#^{k(n-1)}T(2,9))\#(\#^{k(n-1)}T(2,15)).$$

Note that K_3 is a vertex on the path ℓ_3, and as before we obtain the inequality

$$d_n(K_3, \ell_1 \cup \ell_2) \geq 3k/4.$$

If $K = \#^{i(n-1)}T(2,9)$ is any vertex on the path ℓ_1, then Proposition 2.5 implies

$$d_n(K_3, K) \geq \frac{|e_5(K_3) - e_5(K)|}{4(n-1)} = \frac{|4(n-1)k - 0|}{4(n-1)} = k.$$

The above calculation use the facts that $\Sigma_5(T(2,9) \cong \Sigma(2,9,5)$ which is an integral homology sphere, and $\Sigma_5(T(2,15)) \cong \Sigma(2,15,5)$. According to Example 2.10 we obtain $H_1(\Sigma(2,15,5);\mathbb{Z}) \cong (\mathbb{Z}_2)^4$. Thus $d_n(K_3, \ell_1) \geq k$.

Next, we consider $d_n(K_3, \ell_2)$. Let K be a vertex on the path ℓ_2. Then K is of the form $(\#^{2k(n-1)}T(2,9))\#(\#^{i(n-1)}T(2,15))$ where $0 \leq i \leq 2k$. Consider again the 9-fold cyclic covers of S^3 with the branching sets $T(2,9)$ and $T(2,15)$ as before. Then

$$d_n(K_3, K) \geq \frac{|e_9(K) - e_9(K_3)|}{8(n-1)} \geq \frac{6(n-1)k + 2i(n-1)}{8(n-1)} \geq \frac{3k}{4}.$$

Given any $\delta \geq 0$, pick k as large enough so that $3k/4 > \delta$. Then we have constructed a geodesic traingle $\{\ell_1, \ell_2, \ell_3\}$ that is not δ-thin. Accordingly we find again that $\mathcal{K}_{H(n)}$ is not δ-hyperbolic for any $\delta \geq 0$.

4. Concordance Knot Graphs

This section is devoted to the study of concordance knot graphs. The first half of the section studies the graph $\mathcal{CK}_{\times}$ associated to the crossing change operation $\times$, and provides a proof of Part (i, ii) of Theorem 1.3. The second half proves Theorem 1.7.

4.1. **Nonhyperbolicity of the Gordian concordance graph.** For simplicity of notation let $s'(K) = \frac{1}{2}s(K)$, where $s(K)$ is the Rasmussen invariant of the knot K [Ras10]. We shall still refer to s' itself as the Rasmussen invariant. Let $\tau(K)$ denote the Ozsváth-Szabó concordance tau invariant [OS03]. Then the distance function d on $\mathcal{CK}_{\times}$ satisfies the lower bounds

$$d([K],[K']) \geq |\tau(K) - \tau(K')| \quad \text{and} \quad d([K],[K']) \geq |s'(K) - s'(K')|,$$

whenever the knot K and K' have diagrams that differ by a single crossing change [OS03, Ras10].

A result of Hedden–Ording [HO08] stipulates that the knot $K_{0,1} := D_+(T(2,3),2)$ (the 2-twisted positive Whitehead double of the right-handed trefoil knot $T(2,3)$) has Ozsváth-Szabó and Rasmussen invariants given by

$$\tau(K_{0,1}) = 0 \quad \text{and} \quad s'(K_{0,1}) = 1.$$

Since all Whitehead doubles of nontrivial knots have unknotting number equal to 1, we obtain $u(K_{0,1}) = 1$. Let $K_{1,1} = -T(2,3)$ and observe that

$$\tau(K_{1,1}) = -1, \quad s'(K_{1,1}) = 1, \quad \text{and} \quad u(K_{1,1}) = 1.$$

Pick an integer $k \in \mathbb{N}$, and form a triangle with edges ℓ_1, ℓ_2, ℓ_3 constructed as follows:

- The edge ℓ_1 connects the class of the class of the unknot $[U]$ to the class of the knot $[\#^k K_{0,1}]$ with intermediate vertices given by $[\#^m K_{0,1}]$, $m = 1, , \ldots, k-1$. Let $L := \#^k K_{0,1}$.
- The edge ℓ_2 connects $[L] = [\#^k K_{0,1}]$ to $[L\#(\#^k K_{1,1})]$ with intermediate vertices given by $[L\#(\#^m K_{1,1})]$, $m = 1, , \ldots, k-1$.
- The edge ℓ_3 connects $[U]$ to $[L\#(\#^k K_{1,1})]$ with intermediate vertices given by

$$[U] \leftrightarrow [K_{1,1}] \leftrightarrow [\#^2 K_{1,1}] \leftrightarrow [\#^3 K_{1,1}] \leftrightarrow \cdots \leftrightarrow [\#^k K_{1,1}] \leftrightarrow$$

$$[\#^k K_{1,1}] \leftrightarrow [K_{0,1}\#(\#^k K_{1,1})] \leftrightarrow [(\#^2 K_{0,1})\#(\#^k K_{1,1})] \leftrightarrow \cdots \leftrightarrow [(\#^k K_{0,1})\#(\#^k K_{1,1})].$$

We first show that all three edges are geodesic paths in $\mathcal{CK}_{\times}$.

Pick a pair of vertices $[\#^m K_{0,1}]$ and $[\#^n K_{0,1}]$ in ℓ_1, with $m, n \in \{0, \ldots, k\}$. Then $\#^m K_{0,1}$ and $\#^n K_{0,1}$ are related by $|m-n|$ crossing changes, showing that $d([\#^m K_{0,1}], [\#^n K_{0,1}]) \leq |m-n|$. On the other hand

$$d([\#^m K_{0,1}], [\#^n K_{0,1}]) \geq |s'(\#^m K_{0,1}) - s'(\#^n K_{0,1})| = |m-n|,$$

showing that $d([\#^m K_{0,1}], [\#^n K_{0,1}]) = |m-n|$ and thus that ℓ_1 is a geodesic edge.

Similarly, pick a pair of vertices $[L\#(\#^m K_{1,1})]$ and $[L\#(\#^n K_{1,1})]$ in ℓ_2, with $m, n \in \{0, \ldots, k\}$. Then $L\#(\#^m K_{1,1})$ and $L\#(\#^n K_{1,1})$ are related by $|m-n|$ crossing changes, showing that $d([L\#(\#^m K_{1,1})], [L\#(\#^n K_{1,1})]) \leq |m-n|$. On the other hand

$$d([L\#(\#^m K_{1,1})], [L\#(\#^n K_{1,1})]) \geq |\tau(L\#(\#^m K_{1,1})) - \tau(L\#(\#^n K_{1,1}))| = |m-n|,$$

showing that $d([L\#(\#^m K_{1,1})], [L\#(\#^n K_{1,1})]) = |m-n|$ and thus that ℓ_2 is a geodesic edge.

Lastly, consider two vertices from ℓ_3. Since the value of s' is increasing by exactly 1 as we pass from the starting vertex $[U]$ of ℓ_3 towards the final vertex $[L\#(\#^k K_{1,1})]$ of ℓ_3, and since every pair of neighboring vertices in ℓ_3 are related by a crossing change, a similar argument applies here too, showing $\ell_1 \cup \ell_2 \cup \ell_3$ to form a geodesic triangle.

Finally, consider the "midpoint" vertex $M = [\#^k K_{1,1}]$ on ℓ_3. By direct computation we find that

$$d[M, \ell_1] = \min_{0 \le m \le k} d[M, [\#^m K_{0,1}]) \ge \min_{0 \le m \le k} |\tau(\#^k K_{1,1}) - \tau(\#^m K_{0,1})| = k,$$
$$d[M, \ell_2] = \min_{0 \le m \le k} d[M, [L\#(\#^m K_{1,1})]) \ge \min_{0 \le m \le k} |\tau(\#^k K_{1,1}) - \tau(L\#(\#^m K_{1,1}))| = |k - m|,$$
$$d[M, \ell_2] = \min_{0 \le m \le k} d[M, [L\#(\#^m K_{1,1})]) \ge \min_{0 \le m \le k} |s'(\#^k K_{1,1}) - s'(L\#(\#^m K_{1,1}))| = m.$$

Since m ranges from 0 to k, we find that

$$d(M, \ell_1) \ge k \quad \text{and} \quad d(M, \ell_2) \ge k/2.$$

Given any $\delta \ge 0$ and picking $k > 2\delta$ shows that the geodesic triangle $\ell_1 \cup \ell_2 \cup \ell_3$ in $\mathcal{CK}_{\times}$ is not δ-thin, and that therefore $\mathcal{CK}_{\times}$ is not δ-hyperbolic for any $\delta \ge 0$. This proves part (ii) of Theorem 1.3.

Remark 4.1. We were not able to prove that the concordance knot graphs $\mathcal{CK}_{H(n)}$ are not δ-hyperbolic, even for the base case of $n = 2$. This is chiefly because we do not know of two "independent" lower bounds on the metric d on $\mathcal{CK}_{H(n)}$, akin to the roles played by τ and s' for the case of $\mathcal{CK}_{\times}$.

Corollary 4.1. *The knot graph $\mathcal{K}_{\times}$ is not Gromov hyperbolic.*

Proof. It suffices to construct a geodesic triangle which is not δ-thin for any $\delta \ge 0$. We still use the triangle with vertices the unknot U, $\#^k K_{0,1}, L\#(\#^k K_{1,1})$ and edges ℓ_1, ℓ_2, ℓ_3 as above. We first claim it is also a geodesic triangle in the knot graph $\mathcal{K}_{\times}$. Note that for a pair of vertices $\#^m K_{0,1}$ and $\#^n K_{0,1}$ in the edge ℓ_1 with $m, n \in \{0, \cdots, k\}$,

$$d_{\times}(\#^m K_{0,1}, \#^n K_{0,1}) \le |m - n|,$$

where $d_{\times}$ is the induced metric in $\mathcal{K}_{\times}$. On the other hand,

$$d_{\times}(\#^m K_{0,1}, \#^n K_{0,1}) \ge d([\#^m K_{0,1}], [\#^n K_{0,1}]) \ge |m - n|$$

where d is the metric in the concordance graph $\mathcal{CK}_{\times}$. Hence $d_{\times}(\#^m K_{0,1}, \#^n K_{0,1}) = |m - n|$, and ℓ_1 is a geodesic. By using a similar argument, we can prove that ℓ_2, ℓ_3 are also geodesics and the geodesic triangle in $\mathcal{CK}_{\times}$ is also a geodesic triangle in $\mathcal{K}_{\times}$. It is also not hard to see that

$$d_{\times}(\#^k K_{1,1}, \ell_1 \cup \ell_2) \ge k/2$$

by using a similar argument. Hence, $\mathcal{K}_{\times}$ is not δ-hyperbolic for any $\delta \ge 0$, and therefor not Gromov hyperbolic. $\square$

4.2. **Homogeneity of concordance knot graphs.** Recall that a metric space (X, d) is homogeneous if for every $x, y \in X$ there exists an isometry $\psi : X \to X$ with $\psi(x) = y$. The goal of this section is to show that any concordance graph is homogeneous, which proves Part (i) of Theorem 1.7. Toward that goal, let $\mathbb{O} = \{\mathcal{O}_1, \ldots, \mathcal{O}_n\}$ be a collection of distinct unknotting operations, let $\mathcal{CK}_{\mathbb{O}}$ be the associated concordance graph, and let d denote its induced metric. For a fixed pair of knots K, K', let $\psi_{K,K'} : \mathcal{CK}_{\mathbb{O}} \to \mathcal{CK}_{\mathbb{O}}$ be the function

$$\psi([L]) = [L\#(-\overline{K})\#K'],$$

where $-\overline{K}$ denotes the reverse mirror of K. Note that $\psi([K]) = [K']$ and that $\psi_{K,K'}$ is a bijection with inverse $\psi_{K',K}$.

To show that $\psi_{K,K'}$ is an isometry of $\mathcal{CK}_{\mathbb{O}}$, let $[L]$ and $[L']$ be concordance classes with $d([L], [L']) = 1$. Without loss of generality we may assume that the knots L and L' are related by an $\mathcal{O}_i$-move (or its inverse) for some $i \in \{1, \ldots, n\}$. It follows that the knots $L\#(-\overline{K})\#K'$ and $L'\#(-\overline{K})\#K'$ are also related by an $\mathcal{O}_i$-move (or its inverse) showing that

$$d(\psi_{K,K'}([L]), \psi_{K,K'}([L']) = d([L\#(-\overline{K})\#K'], [L'\#(-\overline{K})\#K']) = 1.$$

Iterating this argument one finds that for any pair of concordance classes $[L]$ and $[L']$ (with arbitrary $d([L], [L'])$) the following inequality holds:

$$d(\psi_{K,K'}([L]), \psi_{K,K'}([L'])) \leq d([L], [L']).$$

Repeating the argument for $(\psi_{K,K'})^{-1} = \psi_{K',K}$ one obtains the opposite inequality, showing that $\psi_{K,K'}$ is an isometry, proving part (i) of Theorem 1.7.

5. Quotient Knot Graphs

This section studies quotient knot graphs associated to various choices of knot invariants and unknotting operations.

5.1. **Quotients with respect to a single unknotting operation and knot invariant.** Let $\mathcal{O}$ be an unknotting operation on knot diagrams, and let $\mathcal{I}$ be an integer-valued knot invariant compatible with $\mathcal{O}$ (Definition 1.2). In the next theorem let $|\cdot|$ denote the Euclidean norm on $\mathbb{R}$, as well as its restrictions to various subsets of $\mathbb{R}$.

Theorem 5.1. *Let $\mathcal{I}$ be an integer-valued knot invariant with image $\mathbb{N}$, $\mathbb{N} \cup \{0\}$ or $\mathbb{Z}$, and let $\mathcal{O}$ be an unknotting operation compatible with $\mathcal{I}$. Let $\mathcal{QK}^{\mathcal{I}}_{\mathcal{O}}$ be the associated quotient knot graph and let d denote its metric. If for every $n \in Im(\mathcal{I})$ there exists a knot K_n with $\mathcal{I}(K_n) = n$, and where K_n and K_{n+1} are related by an $\mathcal{O}$-move or its inverse, then the function*

$$\mathcal{I} : (\mathcal{QK}^{\mathcal{I}}_{\mathcal{O}}, d) \to (Im(\mathcal{I}), |\cdot|)$$

is an isometry.

Proof. Recall that for a knot K the equivalence class $[K]^{\mathcal{I}}_{\mathcal{O}}$ consists of all knot K' with $\mathcal{I}(K') = \mathcal{I}(K)$. If $[K]^{\mathcal{I}}_{\mathcal{O}} \neq [K']^{\mathcal{I}}_{\mathcal{O}}$ are any two vertices in $Q\mathcal{K}^{\mathcal{I}}_{\mathcal{O}}$ with $d([K]^{\mathcal{I}}_{\mathcal{O}}, [K']^{\mathcal{I}}_{\mathcal{O}}) = n \geq 1$, let $K_0, \ldots, K_n$ be knots such that a single $\mathcal{O}$-move or its inverse relates K_i to K_{i+1}, and such that $K_0 \in [K]^{\mathcal{I}}_{\mathcal{O}}$ and $K_n \in [K']^{\mathcal{I}}_{\mathcal{O}}$. Write $\mathcal{I}(K_{i+1}) = \mathcal{I}(K_i) + \epsilon_{i+1}$ for some choice of $\epsilon_{i+1} \in \{-1, 0, 1\}$ (which is possible since $\mathcal{O}$ and $\mathcal{I}$ have been assumed to be compatible). Then

$$|\mathcal{I}(K') - \mathcal{I}(K)| = |\mathcal{I}(K_n) - \mathcal{I}(K_0)| = |\epsilon_1 + \cdots + \epsilon_n| \leq n = d([K']^{\mathcal{I}}_{\mathcal{O}}, [K]^{\mathcal{I}}_{\mathcal{O}}).$$

It follows that

$$|n - m| = |\mathcal{I}(K_n) - \mathcal{I}(K_m)| \leq d([K_n]^{\mathcal{I}}_{\mathcal{O}}, [K_m]^{\mathcal{I}}_{\mathcal{O}}).$$

On the other hand, since $K_n \in [K_n]^{\mathcal{I}}_{\mathcal{O}}$ and $K_m \in [K_m]^{\mathcal{I}}_{\mathcal{O}}$ and K_n and K_m differ by at most $|n-m|$ $\mathcal{O}$-moves and/or their inverses, it follows that $d([K_n]^{\mathcal{I}}_{\mathcal{O}}, [K_m]^{\mathcal{I}}_{\mathcal{O}}) \leq |n-m|$ and hence

$$d([K_n]^{\mathcal{I}}_{\mathcal{O}}, [K_m]^{\mathcal{I}}_{\mathcal{O}}) = |n - m|,$$

completing the proof of the theorem. □

Example 5.2. Take $\mathcal{O}$ to be a noncoherent band move (an $H(2)$ move), take $\mathcal{I}$ to be γ_4 (the non-orientable smooth 4-genus) and for $n \in \mathbb{N} = \mathrm{Im}(\mathcal{I})$, let $K_n = T(2n + 2, 2n + 1)$. Then by [Bat14], the knots K_n satisfy the assumptions of Theorem 5.1, leading to the isometry $Q\mathcal{K}^{\gamma_4}_{H(2)} \cong \mathbb{N}$.

Example 5.3. Taking $\mathcal{O}$ to be a crossing change operation $\times$, taking $\mathcal{I} = g_4$ and for $n \in \mathbb{N} \cup \{0\} = \mathrm{Im}(g_4)$ letting $K_n = T(2n+1, 2)$ satisfies the assumptions of Theorem 5.1, giving the isometry $Q\mathcal{K}^{g_4}_{\times} \cong \mathbb{N} \cup \{0\}$.

Example 5.4. Taking $\mathcal{O}$ to be a crossing change operation $\times$, taking $\mathcal{I} = \tau$ (the Ozsváth-Szabó tau invariant [OS03]) and for $n \in \mathbb{Z} = \mathrm{Im}(\tau)$ letting

$$K_n = \begin{cases} T(2n+1, 2) & , \quad n \geq 0, \\ T(2n-1, 2) & , \quad n < 0, \end{cases}$$

satisfies the assumptions of Theorem 5.1. Thus $Q\mathcal{K}^{\tau}_{\times}$ is isometric to $\mathbb{Z}$.

Example 5.5. Taking $\mathcal{O}$ to be a crossing change operation $\times$, letting $\mathcal{I} = s'$ (half of the Rasmussen s invariant [Ras10]) and for $n \in \mathbb{Z} = \mathrm{Im}(s')$ letting

$$K_n = \begin{cases} T(-2n-1, 2) & , \quad n \geq 0, \\ T(-2n+1, 2) & , \quad n < 0, \end{cases}$$

satisfies the assumptions of Theorem 5.1, proving that the spaces $Q\mathcal{K}^{s/2}_{\times}$ and $\mathbb{Z}$ are isometric.

Example 5.6. This example illustrates that when picking an unknotting operation $\mathcal{O}$ and knot invariant $\mathcal{I}$ that are not compatible in the sense of Definition 1.2, the resulting metric space $\mathcal{QK}^{\mathcal{I}}_{\mathcal{O}}$ can be very different from what is asserted in Theorem 5.1.

Take $\mathcal{O}$ to be the operation of noncoherent band moves ($H(2)$ moves) and let $\mathcal{I} = g_4$. Note that the values of g_4 of a pair of knots differing by a single $H(2)$-move may differ by an arbitrarily large amount. The vertices of $\mathcal{QK}^{g_4}_{H(2)}$ can be identified with $\mathbb{N} \cup \{0\} = \mathrm{Im}(g_4)$, under the correspondence $n \mapsto [T(2n+1,2)]^{g_4}_{H(2)}$. For each $n \in \mathbb{N}$ a single band move renders $T(2n+1,2)$ unknotted, showing that

$$d(n,0) = 1, \quad \forall n \in \mathbb{N},$$

where d is the induced metric on $\mathcal{QK}^{g_4}_{H(2)}$. In particular, $d(n,m) \leq 2$ for all $n, m \in \mathbb{N} \cup \{0\}$. There is a band move that transforms $T(2n+1,2)$ into $T(2n-3,2)$ (see for example [LMV19, Figure 2]) showing additionally that

$$d(n,m) = 1, \quad \text{if } |n-m| = 4.$$

These relations don't fully pin down the metric space $\mathcal{QK}^{g_4}_{H(2)}$ but they show that it is not isometric to a subspace of $\mathbb{R}$.

Theorem 5.1 in conjunction with Examples 5.2–5.5, proves Theorem 1.4.

5.2. **Quotients with respect to a single operation and two knot invariants.** Let $||\cdot||_1$ and $||\cdot||_\infty$ denote the ℓ_1- and ℓ_∞-norms on $\mathbb{R}^2$, as well as their restrictions to various subsets of $\mathbb{R}^2$.

Theorem 5.7. *Let $\mathcal{O}$ be an unknotting operation and let $\mathcal{I}_1, \mathcal{I}_2$ be two integer-valued knot invariants compatible with $\mathcal{O}$. Assume that if (m_1, n_1) and (m_2, n_2) both lie in the image of $\mathcal{I}_1 \times \mathcal{I}_2$, then either*

(1)
$$\cup_{m,n} \{(m_1, n), (m, n_2\} \subset Im(\mathcal{I}_1 \times \mathcal{I}_2) \quad \text{or} \quad \cup_{m,n} \{(m, n_1), (m_2, n\} \subset Im(\mathcal{I}_1 \times \mathcal{I}_2),$$

with both unions taken over integers m between m_1 and m_2, and integers n between n_1 and n_2. Let $\mathbb{I} = \{\mathcal{I}_1, \mathcal{I}_2\}$ and let $\mathcal{QK}^{\mathbb{I}}_{\mathcal{O}}$ be the associated quotient knot graph with metric d. Suppose there exists a family $\{K_{m,n} \,|\, (m,n) \in Im(\mathcal{I}_1 \times \mathcal{I}_2)\}$ of distinct knots $K_{m,n}$ such that $\mathcal{I}_1(K_{m,n}) = m$, $\mathcal{I}_2(K_{m,n}) = n$, and such that

- *$K_{m,n}$ and $K_{m,n+1}$ are related by an $\mathcal{O}$-move or its inverse, whenever (m,n) and $(m,n+1)$ both lie in the image of $\mathcal{I}_1 \times \mathcal{I}_2$, and*
- *$K_{m,n}$ and $K_{m+1,n}$ are related by an $\mathcal{O}$-move or its inverse, whenever (m,n) and $(m+1,n)$ both lie in the image of $\mathcal{I}_1 \times \mathcal{I}_2$.*

Then the function

$$\mathcal{I}_1 \times \mathcal{I}_2 : (\mathcal{QK}^{\mathbb{I}}_{\mathcal{O}}, d) \to (Im(\mathcal{I}_1 \times \mathcal{I}_2), |\cdot|_1)$$

is bi-Lipschitz and satisfies the inequality

$$||(m_1, n_1),\ (m_2, n_2)||_\infty \leq d([K_{m_1,n_1}], [K_{m_2,n_2}]) \leq ||(m_1, n_1),\ (m_2, n_2)||_1.$$

Proof. Given a knot K, for simplicity of notation we shall write $[K]$ to mean the equivalence class $[K]^{\mathbb{I}}_{\mathcal{O}}$. The compatibility assumption between $\mathcal{O}$ and $\mathbb{I}$ implies that

$$d([K], [K']) \geq |\mathcal{I}_j(K) - \mathcal{I}_j(K')|,$$

for $j = 1, 2$ and for any pair of knots K, K' related by an $\mathcal{O}$-move or its inverse. From these, in complete analogy with the proof of Theorem 5.1 (while relying on assumption (1)), one obtains

$$d([K_{m,n_1}], [K_{m,n_2}]) = |n_1 - n_2| \quad \text{and} \quad d([K_{m_1,n}], [K_{m_2,n}]) = |m_1 - m_2|,$$

whenever $(m, n_1), (m, n_2), (m_1, n), (m_2, n)$ lie in the image of $\mathcal{I}_1 \times \mathcal{I}_2$. These two equalities show that there are no $\mathcal{O}$-moves between knots K_{m,n_1} and K_{m,n_2} if $|n_1 - n_2| \geq 2$, and similarly there are no $\mathcal{O}$-moves between knots $K_{m_1,n}$ and $K_{m_2,n}$ if $|m_1 - m_2| \geq 2$.

Suppose there is an $\mathcal{O}$-move from K_{m_1,n_1} to K_{m_2,n_2} for $m_1 \neq m_2$ and $n_1 \neq n_2$. Then

$$1 = d([K_{m_1,n_1}], [K_{m_2,n_2}]) \geq |\mathcal{I}_1(K_{m_1,n_1}) - \mathcal{I}_1(K_{m_2,n_2})| = |m_1 - m_2|,$$
$$1 = d([K_{m_1,n_1}], [K_{m_2,n_2}]) \geq |\mathcal{I}_2(K_{m_1,n_1}) - \mathcal{I}_2(K_{m_2,n_2})| = |n_1 - n_2|.$$

We find that the only such $\mathcal{O}$-moves possible are the ones connecting a knot $K_{m,n}$ to the knots $K_{m\pm1,n\pm1}$ (with both signs chosen arbitrarily). Observe then that the distance $d([K_{m_1,n_1}], [K_{m_2,n_2}])$ is minimized when all possible $\mathcal{O}$-moves of these types exist. Thus, a lower bound on $d([K_{m_1,n_1}], [K_{m_2,n_2}])$ is given by

$$|(m_1, n_1), (m_2, n_2)|_\infty = \max\{|m_1 - m_2|,\ |n_1 - n_2|\} \leq d([K_{m_1,n_1}], [K_{m_2,n_2}]).$$

On the other hand, for an arbitrary pair $(m_1, n_1), (m_2, n_2) \in \mathrm{Im}(\mathcal{I}_1 \times \mathcal{I}_2)$, it is clear that

$$d([K_{m_1,n_1}], [K_{m_2,n_2}]) \leq |n_1 - n_2| + |m_1 - m_2| = ||(m_1, n_1), (m_2, n_1)||_1.$$

This claim relies on assumption (1). Indeed, if for instance $\cup_{m,n}\{(m_1, n), (m, n_2)\} \subset \mathrm{Im}(\mathcal{I}_1 \times \mathcal{I}_2)$ (with m between m_1 and m_2 and n between n_1 and n_2), then there are $|n_1 - n_2|$ $\mathcal{O}$-moves that connect K_{m_1,n_1} to K_{m_1,n_2}, and a further $|m_1 - m_2|$ $\mathcal{O}$-moves that connect the latter knot to K_{m_2,n_2}. A similar argument applies in the case that $\cup_{m,n}\{(m, n_1), (m_2, n\} \subset \mathrm{Im}(\mathcal{I}_1 \times \mathcal{I}_2)$. □

Example 5.8. Consider $\mathcal{O}$ to be the crossing change operation $\times$, and let $\mathcal{I}_1 = g_4$ and $\mathcal{I}_2 = u$ (where $u(K)$ is the unknotting number of the knot K). Since for any knot K one has the bound $g_4(K) \leq u(K)$, it follows that the image of $g_4 \times u$ is a subset of the second octant of $\mathbb{Z}^2$. As we shall see, the image is actually equal to said octant.

Let $K_{0,1}$ and $K_{1,1}$ be the knots

$$K_{0,1} = 6_1 = \text{Stevedors knot},$$
$$K_{1,1} = 3_1 = \text{Trefoil knot}.$$

It is well known and easy to verify that $g_4(K_{0,1}) = 0$, $u(K_{0,1}) = 1$, $g_4(K_{1,1}) = 1 = u(K_{1,1})$. For integers $0 \leq m \leq n$ define $K_{m,n}$ as

$$K_{m,n} = (\#^{n-m} K_{0,1})\#(\#^m K_{1,1}).$$

In the above $\#^0 K$ denotes the unknot. Observe that

$$\begin{aligned} g_4(K_{m,n}) &= g_4((\#^{n-m} K_{0,1})\#(\#^m K_{1,1})) \\ &\leq (n-m)g_4(K_{0,1}) + mg_4(K_{1,1}) \\ &= m, \end{aligned}$$

and

$$\begin{aligned} u(K_{m,n}) &= u((\#^{n-m} K_{0,1})\#(\#^m K_{1,1})) \\ &\leq (n-m)u(K_{0,1}) + mu_4(K_{1,1}) \\ &= n. \end{aligned}$$

Since

$$\tau(K_{m,n}) = (n-m)\tau(K_{0,1}) + m\tau(K_{1,1}) = m$$

and since $|\tau(K)| \leq g_4(K)$ for any knot K, it follows that $g_4(K_{m,n}) = m$. Recall that there is a lower bound for the unknotting number given by the minimal number of generators of $H_1(\Sigma(K);\mathbb{Z})$ (see [Wen37, page 690] or [Nak81]). That is,

$$e_2(K) \leq u(K).$$

For $K_{m,n}$ one finds that

$$\begin{aligned} H_1(\Sigma(K_{m,n});\mathbb{Z}) &= \left(\oplus_{i=1}^{n-m} H_1(L(9,7);\mathbb{Z})\right) \oplus \left(\oplus_{j=1}^{m} H_1(L(3,1);\mathbb{Z})\right) \\ &= (\mathbb{Z}_9)^{n-m} \oplus (\mathbb{Z}_3)^m. \end{aligned}$$

The minimal number of generators for this homology group is n, implying that $u(K_{m,n}) = n$, and in particular that

$$\mathrm{Im}(g_4 \times u) = \{(m,n) \in \mathbb{Z}^2 \,|\, 0 \leq m \leq n\}.$$

This shows that condition (1) from Theorem 5.7 applies in the current setting.

Lastly, the knot $K_{m,n+1} = (\#^{n+1-m} K_{0,1})\#(\#^m K_{1,1})$ is related to the knot $K_{m,n} = (\#^{n-m} K_{0,1})\#(\#^m K_{1,1})$ via a crossing change that unknots one of the $K_{0,1}$ summands of $K_{m,n+1}$. Similarly $K_{m+1,n}$ is related to $K_{m,n}$ via the crossing change that unknots one of the $K_{1,1}$ summands of $K_{m+1,n}$.

It follows then from Theorem 5.7 that the function

$$g_4 \times u : (Q\mathcal{K}_{\times}^{g_4,u}, d) \to (\mathrm{Im}(g_4 \times u, |\cdot|_1)$$

is bi-Lipschitz, and in particular, $(Q\mathcal{K}_{\times}^{g_4,u}, d)$ is not δ-hyperbolic for any $\delta \geq 0$. This proves part (iii) of Theorem 1.3, which completes the proof of this theorem.

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